

## COLES POINT, Australia

WMO: 946580

Lat: 34.375S

Lon: 135.374E

Elev: 94

StdP: 14.65

Time Zone: 9.50 (AUA)

Period: 94-17

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 40.6       | 42.3 | 30.2                          | 24.4 | 73.6 | 34.0 | 28.7 | 65.3 | 31.6                  | 54.2 | 29.5 | 54.8 | 9.3                   | 40   | 0.586 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 18.2                   | 94.2            | 65.5 | 89.9 | 64.4 | 86.0 | 63.5 | 69.6                | 82.5 | 67.8 | 80.8 | 66.2 | 78.8 | 12.8                 | 330  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.1                            | 96.6 | 71.9 | 63.9 | 89.2 | 70.8 | 62.1 | 83.7 | 69.2 | 33.7          | 82.3 | 32.2 | 80.7 | 30.9 | 78.7 | 78.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 26.4          | 23.5 | 21.3 | DB                         | 37.2 | 102.0              | 1.5 | 2.9                                                | 36.1 | 104.1      | 35.2 | 105.8      | 34.4 | 107.5      | 33.3 | 109.6 | (4) |
| (5)               |      |      | WB                         | 36.3 | 72.5               | 1.6 | 2.4                                                | 35.1 | 74.2       | 34.2 | 75.5       | 33.3 | 76.9       | 32.1 | 78.6  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 60.9        | 69.3  | 69.7  | 66.4  | 62.3  | 57.7  | 54.1  | 52.6  | 53.3  | 56.2  | 59.3  | 63.9  | 66.9  | (6)  |  |
|                                                                        | DBStd       | 8.05        | 6.91  | 6.79  | 6.12  | 4.80  | 3.77  | 3.38  | 3.07  | 3.20  | 4.46  | 5.81  | 6.88  | 6.80  | (7)  |  |
|                                                                        | HDD50       | 24          | 0     | 0     | 0     | 0     | 0     | 6     | 10    | 6     | 2     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD65       | 2129        | 21    | 17    | 50    | 111   | 232   | 327   | 386   | 361   | 269   | 199   | 104   | 52    | (9)  |  |
|                                                                        | CDD50       | 4010        | 600   | 552   | 509   | 368   | 237   | 129   | 89    | 110   | 186   | 289   | 417   | 524   | (10) |  |
|                                                                        | CDD65       | 639         | 155   | 149   | 94    | 29    | 4     | 0     | 0     | 4     | 23    | 70    | 111   |       | (11) |  |
|                                                                        | CDH74       | 6106        | 1563  | 1372  | 751   | 159   | 27    | 1     | 0     | 1     | 43    | 270   | 771   | 1148  | (12) |  |
|                                                                        | CDH80       | 2512        | 732   | 588   | 276   | 36    | 4     | 0     | 0     | 7     | 76    | 301   | 492   |       | (13) |  |
| (14) Wind                                                              | WSAvg       | 11.6        | 12.2  | 11.9  | 10.9  | 10.1  | 10.9  | 11.6  | 12.3  | 11.9  | 11.9  | 12.2  | 11.9  | 12.0  | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 20.8        | 0.5   | 0.6   | 0.7   | 1.5   | 2.5   | 3.2   | 3.7   | 3.2   | 2.2   | 1.5   | 0.8   | 0.8   | (15) |  |
|                                                                        | PrecMax     | 34.8        | 2.3   | 3.5   | 4.0   | 6.1   | 6.1   | 7.2   | 7.1   | 6.2   | 6.1   | 4.1   | 2.9   | 3.1   | (16) |  |
|                                                                        | PrecMin     | 12.4        | 0.0   | 0.0   | 0.0   | 0.1   | 0.6   | 0.9   | 1.1   | 0.7   | 0.4   | 0.2   | 0.0   | 0.0   | (17) |  |
|                                                                        | PrecStd     | 4.5         | 0.4   | 0.7   | 0.8   | 1.4   | 1.2   | 1.4   | 1.6   | 1.3   | 1.3   | 0.9   | 0.7   | 0.7   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 99.8  | 98.0  | 93.0  | 84.4  | 77.6  | 67.4  | 65.3  | 70.5  | 79.3  | 87.2  | 93.6  | 97.1  | (19) |  |
|                                                                        |             | MCWB        | 66.5  | 66.7  | 63.9  | 60.7  | 57.6  | 57.3  | 55.1  | 56.9  | 58.8  | 60.4  | 63.6  | 64.1  | (20) |  |
|                                                                        | 2%          | DB          | 93.4  | 93.1  | 87.2  | 78.2  | 70.4  | 63.5  | 61.7  | 65.7  | 73.0  | 81.3  | 87.6  | 90.8  | (21) |  |
|                                                                        |             | MCWB        | 66.2  | 66.9  | 63.8  | 59.4  | 57.5  | 56.3  | 54.8  | 55.2  | 57.7  | 58.9  | 62.6  | 63.4  | (22) |  |
|                                                                        | 5%          | DB          | 88.4  | 87.3  | 82.3  | 74.0  | 66.6  | 61.6  | 59.9  | 62.1  | 68.1  | 76.0  | 83.0  | 85.3  | (23) |  |
|                                                                        |             | MCWB        | 64.7  | 65.7  | 62.8  | 59.3  | 57.2  | 55.9  | 54.5  | 54.6  | 56.6  | 58.1  | 61.4  | 62.2  | (24) |  |
|                                                                        | 10%         | DB          | 82.8  | 82.1  | 77.3  | 70.5  | 64.3  | 60.1  | 58.5  | 59.8  | 64.4  | 70.8  | 77.6  | 80.2  | (25) |  |
|                                                                        |             | MCWB        | 63.5  | 64.6  | 62.1  | 58.9  | 57.0  | 55.1  | 53.9  | 53.9  | 56.1  | 56.9  | 60.0  | 62.0  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 71.5  | 73.5  | 70.4  | 65.0  | 62.8  | 60.8  | 58.9  | 59.7  | 62.1  | 63.9  | 67.5  | 69.9  | (27) |  |
|                                                                        |             | MCDB        | 85.9  | 87.1  | 80.7  | 71.9  | 66.8  | 63.8  | 60.8  | 65.4  | 70.9  | 76.3  | 84.6  | 77.7  | (28) |  |
|                                                                        | 2%          | WB          | 69.2  | 69.7  | 67.5  | 63.5  | 61.2  | 58.5  | 57.3  | 57.4  | 59.9  | 61.6  | 64.9  | 66.9  | (29) |  |
|                                                                        |             | MCDB        | 83.6  | 83.8  | 77.3  | 69.2  | 65.4  | 61.1  | 59.4  | 61.6  | 67.6  | 72.6  | 81.0  | 79.1  | (30) |  |
|                                                                        | 5%          | WB          | 67.2  | 67.7  | 65.6  | 62.3  | 59.8  | 57.3  | 55.8  | 56.1  | 58.4  | 60.0  | 63.1  | 64.9  | (31) |  |
|                                                                        |             | MCDB        | 81.1  | 81.0  | 75.3  | 68.4  | 63.9  | 60.2  | 58.6  | 60.3  | 64.7  | 69.6  | 76.5  | 78.5  | (32) |  |
|                                                                        | 10%         | WB          | 65.4  | 66.1  | 63.9  | 61.0  | 58.4  | 56.0  | 54.6  | 54.8  | 57.0  | 58.5  | 61.5  | 63.3  | (33) |  |
|                                                                        |             | MCDB        | 79.4  | 78.2  | 73.7  | 67.4  | 62.7  | 59.2  | 57.9  | 59.0  | 62.9  | 67.4  | 73.8  | 76.6  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 19.0  | 18.2  | 16.9  | 15.7  | 13.4  | 12.0  | 11.8  | 13.3  | 15.1  | 17.9  | 18.7  | 18.8  | (35) |  |
|                                                                        |             | MCDBR       | 28.2  | 25.5  | 23.8  | 20.9  | 17.7  | 13.7  | 13.8  | 18.5  | 23.3  | 27.5  | 27.6  | 27.9  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 9.5   | 8.8   | 8.4   | 9.1   | 9.8   | 9.2   | 9.7   | 11.4  | 11.8  | 11.2  | 10.0  | 9.7   | (37) |  |
|                                                                        |             | MCWBR       | 22.4  | 20.6  | 18.3  | 15.5  | 13.2  | 11.0  | 11.4  | 14.8  | 18.0  | 21.2  | 22.8  | 21.9  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.343 | 0.342 | 0.327 | 0.320 | 0.311 | 0.305 | 0.299 | 0.308 | 0.336 | 0.340 | 0.344 | 0.335 | (40) |  |
|                                                                        |             | taud        | 2.552 | 2.567 | 2.599 | 2.610 | 2.616 | 2.619 | 2.632 | 2.588 | 2.492 | 2.484 | 2.497 | 2.552 | (41) |  |
|                                                                        | Ebn at Noon |             | 315   | 309   | 302   | 287   | 272   | 266   | 274   | 287   | 294   | 306   | 313   | 319   | (42) |  |
|                                                                        |             | Edh at Noon | 34    | 33    | 30    | 26    | 23    | 21    | 22    | 26    | 32    | 35    | 36    | 35    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 2424  | 2133  | 1728  | 1253  | 880   | 733   | 803   | 1090  | 1498  | 1913  | 2222  | 2399  | (44) |  |
|                                                                        | RadStd      |             | 104   | 104   | 77    | 77    | 41    | 35    | 37    | 68    | 100   | 125   | 128   | 106   | (45) |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (10 neighbors) | +0.52 | N/A     | N/A    | +1.66   | N/A   | N/A   | N/A         | -184  | +218  | N/A   |

Nomenclature: See separate page